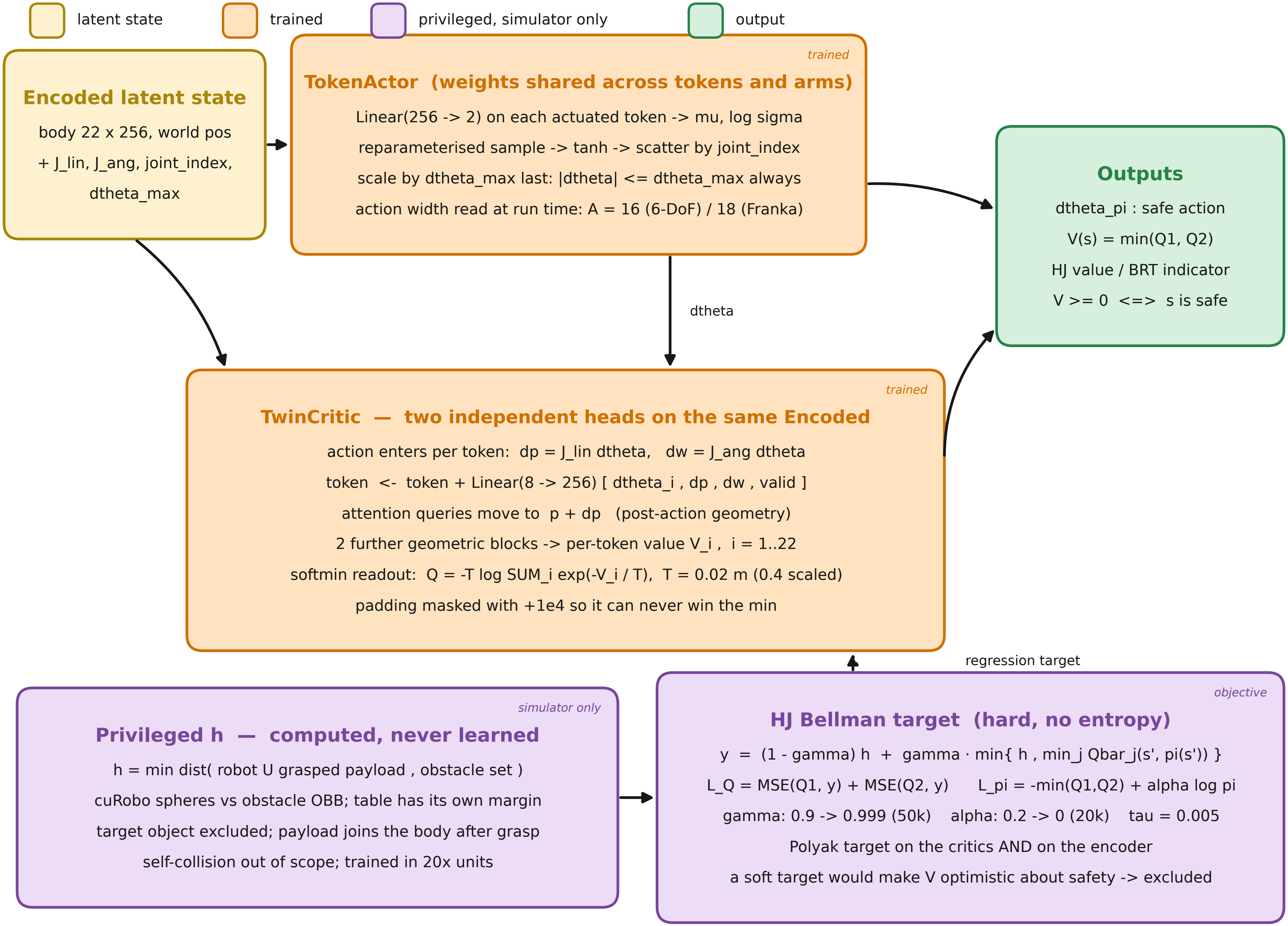
latent state
trained
privileged, simulator only
output
Encoded latent state
body 22 x 256, world pos
+ J_lin, J_ang, joint_index,
dtheta_max
trained
TokenActor (weights shared across tokens and arms)
Linear(256 -> 2) on each actuated token -> mu, log sigma
reparameterised sample -> tanh -> scatter by joint_index
scale by dtheta_max last: |dtheta| <= dtheta_max always
action width read at run time: A = 16 (6-DoF) / 18 (Franka)
Outputs
dtheta_pi : safe action
V(s) = min(Q1, Q2)
HJ value / BRT indicator
V >= 0 <=> s is safe
dtheta
trained
TwinCritic — two independent heads on the same Encoded
action enters per token: dp = J_lin dtheta, dw = J_ang dtheta
token <- token + Linear(8 -> 256) [dtheta_i , dp , dw , valid]
attention queries move to p + dp (post-action geometry)
2 further geometric blocks -> per-token value V_i , i = 1..22
softmin readout: Q = -T log SUM_i exp(-V_i / T), T = 0.02 m (0.4 scaled)
padding masked with +1e4 so it can never win the min
regression target
simulator only
Privileged h — computed, never learned
h = min dist(robot U grasped payload , obstacle set)
cuRobo spheres vs obstacle OBB; table has its own margin
target object excluded; payload joins the body after grasp
self-collision out of scope; trained in 20x units
objective
HJ Bellman target (hard, no entropy)
y = (1 - gamma) h + gamma · min{ h , min_j Qbar_j(s', pi(s')) }
L_Q = MSE(Q1, y) + MSE(Q2, y) L_pi = -min(Q1,Q2) + alpha log pi
gamma: 0.9 -> 0.999 (50k) alpha: 0.2 -> 0 (20k) tau = 0.005
Polyak target on the critics AND on the encoder
a soft target would make V optimistic about safety -> excluded